Product Optimization & Revenue Contribution Analysis for Afficionado Coffee Roasters

1. Introduction

In the specialty coffee retail industry, product performance varies significantly across categories and locations. Afficionado Coffee Roasters aims to improve its decision-making by understanding which products drive revenue and which contribute minimally. This project focuses on analyzing transaction-level data to uncover product-level insights and optimize the overall menu.

2. Dataset Description

The dataset consists of transaction-level records for the year 2025 with the following attributes:

transaction_id : Unique identifier for each transaction
year: Year of transaction (2025)
transaction_time: Time of purchase (HH:MM:SS)
transaction_qty: Number of items purchased
store_id: Store identifier
store_location: Store branch (e.g., Lower Manhattan)
product_id: Unique product identifier
unit_price: Price per unit
product_category: Category (Coffee, Tea, Drinking Chocolate)
product_type: Product variant
product_detail: Specific product description (e.g., Ethiopia Rg, Dark Chocolate Lg)

3. Data Processing

Revenue was calculated using:

Revenue = transaction_qty × unit_price

Time-based features such as hour, weekday, and month were extracted
Data was grouped by:
Product level
Category level
Store location

4. Exploratory Data Analysis (EDA)

4.1 Product Performance

Products were ranked based on total quantity sold and total revenue

High-performing products such as premium coffee blends showed strong revenue contribution

Some products had high volume but lower revenue due to lower pricing

4.2 Category Analysis

Coffee emerged as the dominant category in terms of revenue

Tea contributed moderate revenue with consistent demand

Drinking Chocolate showed lower volume but relatively higher pricing

4.3 Sales Trends

Peak sales were observed during morning hours, indicating strong breakfast demand

Weekday sales were higher compared to weekends

Monthly trends showed consistent demand with minor fluctuations

4.4 Popularity vs Revenue Analysis

A scatter plot revealed:

Some products are highly popular but generate less revenue

Premium products generate higher revenue despite lower sales volume

5. Key Insights

A small group of products contributes to the majority of total revenue (Pareto Principle)

Coffee category is the primary revenue driver across stores

Certain low-priced products generate high sales but limited revenue impact

Several products contribute minimally and can be considered for removal

Revenue concentration indicates dependence on a few key products

6. Recommendations

Focus on promoting high-revenue products (premium coffee blends)

Reduce or redesign low-performing products to simplify the menu

Introduce pricing strategies for high-demand but low-revenue items

Optimize product placement and marketing for “hero” products

Monitor store-level performance to identify location-based preferences

7. Conclusion

This analysis demonstrates that product-level insights are essential for optimizing retail performance. By focusing on high-impact products and minimizing inefficiencies, Afficionado Coffee Roasters can enhance profitability, streamline operations, and improve customer experience.